# Causal Inference with LLMs

## Day 2 — Identification & Estimation


Moses Boudourides

*Faculty, Graduate Program on Data Science*
*Northwestern University*

Moses.Boudourides@northwestern.edu
Moses.Boudourides@gmail.com


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# Day 2 — Identification & Estimation

- From Identification to Estimation
- Estimators
- Adjustment and Weighting
- LLMs and Measurement Error

# From Identification to Estimation

**Two distinct steps in causal inference:**

- **Identification:**
  - Express a causal quantity in terms of observable quantities
  - Example: under ignorability, the ATE $\tau = \mathbb{E}[Y(1) - Y(0)]$ is identified from $P(Y, D, X)$
- **Estimation:**
  - Approximate the identified quantity from a finite sample
  - Specific estimators (regression, matching, IPW) are introduced in the frames that follow

**Key point:**

- Identification is theoretical; estimation is statistical

# Identification vs Estimation (Formal Distinction)

**Identification:**

- A causal quantity $Q$ is identifiable if it can be uniquely expressed as a function of the observed data distribution

**Notation:**

- $V$: collection of all observed variables (e.g., $V = (Y, D, X)$)
- $P(V)$: joint probability distribution of the observed variables

**Formal idea:**

- If two models imply the same observed distribution $P(V)$, they must imply the same value of $Q$ (because the causal quantity $Q$ is uniquely determined by the observable distribution)

**Implication:**

- Identification depends on assumptions, not on sample size

**Note (LLM setting):**

- In practice, we observe $\tilde{V} = (\tilde{Y}, D, X)$, not $V = (Y, D, X)$
- Identification results assume $P(V)$; we only have access to $P(\tilde{V})$
- This gap is the source of measurement error bias (addressed in Part 2)

# Estimand, Estimator, Estimate

**Estimand:**

- The target causal quantity in the population (e.g., $\tau = \mathbb{E}[Y(1) - Y(0)]$)

**Estimator:**

- A rule that maps observed data $\{(Y_i, D_i, X_i)\}_{i=1}^n$ to a numerical value

**Estimate:**

- The realized value of the estimator for a given sample

**Distinction:**

- Estimand: population quantity
- Estimator: mathematical rule
- Estimate: number computed from data

# Notation Guide

**Superscripts and decorations:**

| Symbol | Meaning |
|---|---|
| $\tau$ | True (population) causal quantity — the estimand |
| $\hat{\tau}$ | Estimated value computed from data — the estimate |
| $\tilde{Y}_i$ | LLM-measured (noisy) version of outcome $Y_i$ |
| $\hat{m}_d(x)$ | Estimated conditional expectation function |
| $\hat{e}(X_i)$ | Estimated propensity score |

**Key distinction:**

- $\hat{\cdot}$ (hat): estimated from data — a statistical approximation
- $\tilde{\cdot}$ (tilde): measured by LLM — a noisy proxy for the true variable
- No decoration: true population quantity or true latent variable

# Data Structure for Estimation

**Observed data:**

$$\{(Y_i, D_i, X_i)\}_{i=1}^n$$

**Notation:**

- $i = 1, \ldots, n$: index of units
- $n$: sample size (number of observed units)
- $(Y_i, D_i, X_i)$: observed outcome, treatment, covariates for unit $i$

**Goal:**

- Estimate the ATE $\tau$ from observed data to obtain an estimator $\hat{\tau}$

# Sampling Variability and Finite Samples

**Observation:**

- The estimator $\hat{\tau}$ depends on the particular sample $\{(Y_i, D_i, X_i)\}_{i=1}^n$

**Key distinction:**

- **Systematic bias:** due to confounding, selection, or measurement error
- **Random variability:** due to finite sample size

**Implication:**

- Even if identification holds (no systematic bias), estimates may differ from $\tau$
- The discrepancy decreases as $n$ increases

**Interpretation:**

- Causal inference requires controlling both:
  - bias (via assumptions)
  - variability (via statistical estimation)

# Unbiasedness of Estimators

**Definition:**

- An estimator $\hat{\tau}$ is **unbiased** if

$$\mathbb{E}[\hat{\tau}] = \tau$$

**Interpretation:**

- On average over repeated samples, the estimator recovers the true ATE

**Important distinction:**

- Unbiasedness does not imply accuracy in a given sample
- Variability may still be large when $n$ is small

**Implication:**

- Both unbiasedness and low variability are required for reliable estimation

# Difference-in-Means Estimator

$$\hat{\tau}_{\text{diff}} = \bar{Y}_1 - \bar{Y}_0$$

**Notation:**

- $\bar{Y}_1 = \frac{1}{n_1} \sum_{i:D_i=1} Y_i$: average outcome among treated units
- $\bar{Y}_0 = \frac{1}{n_0} \sum_{i:D_i=0} Y_i$: average outcome among control units
- $n_1 = \sum_{i=1}^{n} D_i, \quad n_0 = \sum_{i=1}^{n} (1 - D_i)$

**Interpretation:**

- Compares average outcomes between treated and control groups

**Key point:**

- Unbiased for the ATE $\tau$ under random assignment of treatment

**Assumes:**

1. Random assignment of treatment
2. SUTVA (no interference between units, single version of treatment)
3. Perfect measurement: $Y_i$ is observed directly (without LLM-induced error)

- Note: assumption 3 fails when $Y_i$ is measured by an LLM

# Sampling Variance of the Difference-in-Means

$$\mathrm{Var}(\hat{\tau}_{\text{diff}}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n}$$

**Notation:**

- $S_1^2$: variance of potential outcomes $Y(1)$ in the population
- $S_0^2$: variance of potential outcomes $Y(0)$ in the population
- $S_\tau^2$: variance of individual treatment effects $Y(1) - Y(0)$

**Interpretation:**

- Variability arises from heterogeneity in outcomes and treatment effects

**Key point:**

- Sampling variability persists even under perfect randomization

# Why Adjustment is Needed

**Observation:**

- The difference-in-means estimator compares groups defined by $D_i = 1$ (under treatment) and $D_i = 0$ (without treatment)

**Problem:**

- If treatment is not randomly assigned, these groups may differ systematically in $X_i$

**Consequence:**

- Differences in outcomes may reflect both:
  - causal effects of $D_i$
  - pre-existing differences in $X_i$

**Goal:**

- Adjust for $X_i$ to isolate the causal effect of $D_i$ on $Y_i$

# Regression Adjustment

**Why regression is needed:**

- Treatment groups ($D_i = 1$ vs $D_i = 0$) may differ in covariates $X_i$
- Direct comparison of outcomes mixes causal effects with differences due to $X_i$

**Idea:**

- Adjust for $X_i$ by modeling the outcome conditional on treatment and covariates

**Specification:**

$$Y_i = \alpha + \tau D_i + \beta^\top X_i + \varepsilon_i$$

**Notation:**

- $\alpha$: intercept (baseline outcome)
- $\tau$: coefficient measuring the effect of treatment
- $\beta$: vector of coefficients for covariates $X_i$
- $\varepsilon_i$: error term capturing unexplained variation

**Interpretation:**

- $\tau$ measures the average difference in outcomes between treated and control units, holding $X_i$ fixed

# Conditions for Valid Regression Adjustment

**Key requirement:**

- The regression model must correctly adjust for confounding variables $X_i$

**Interpretation:**

- After conditioning on $X_i$, the remaining variation in $D_i$ is as good as random

**Connection to ignorability:**

- Validity requires:

$$(Y(1), Y(0)) \perp D \mid X$$

**Implication:**

- If relevant confounders are omitted from $X_i$, the estimate of $\tau$ will be biased

# Propensity Score as a Balancing Score

$$e(X_i) = P(D_i = 1 \mid X_i)$$

**Definition:**

- The propensity score is the probability that unit $i$ receives treatment given its covariates $X_i$

**Balancing property:**

- Conditional on $e(X_i)$, the distribution of $X_i$ is the same in treated and control groups

**Interpretation:**

- Units with the same propensity score are comparable, even if their covariates differ

**Implication:**

- Matching or weighting on $e(X_i)$ removes systematic differences in $X_i$

# Conditional Expectation as a Regression Function

$$m_d(x) := \mathbb{E}[Y \mid D = d, X = x]$$

**Interpretation:**

- $m_d(x)$ is the average outcome among units with:
  - treatment status $D = d$
  - covariates equal to $X = x$

**Key properties:**

- $m_1(x) - m_0(x)$: **conditional average treatment effect (CATE)** at $X = x$
- Averaging over $X$: $\tau = \mathbb{E}_X[m_1(X) - m_0(X)]$

**Example:**

- $W_i$ = speech text; $Y_i = f(W_i)$ sentiment score derived from speech text $W_i$; $D_i$ = party (R=1, D=0);
- $m_1(x)$: expected sentiment for a Republican with covariates $X = x$
- $m_0(x)$: expected sentiment for a Democrat with covariates $X = x$

**Measured version (LLM):**

- $\tilde{m}_d(x) = \mathbb{E}[\tilde{Y}_i \mid D_i = d, X_i = x]$
- In practice we estimate $\tilde{m}_d(x)$, not $m_d(x)$

**Role:**

- Central object in regression-based estimation

# Plug-in Estimator of the ATE

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^{n} (\hat{m}_1(X_i) - \hat{m}_0(X_i))$$

**Notation:**

- $\hat{m}_d(x)$: estimate of $m_d(x)$ obtained from data (e.g., via regression)
- $X_i$: observed covariates of unit $i$
- $\hat{m}_1(X_i)$: predicted outcome for unit $i$ under treatment
- $\hat{m}_0(X_i)$: predicted outcome for unit $i$ under control
- $\hat{m}_1(X_i) - \hat{m}_0(X_i)$: estimated conditional treatment effect at $X_i$

**Overall interpretation:**

- Evaluate the estimated conditional treatment effect at each observed $X_i$
- Average across units to obtain an estimate of $\tau$

# Plug-in Estimator and Regression

**Observation:**

- The plug-in estimator requires estimating the functions

$$m_d(x) = \mathbb{E}[Y \mid D = d, X = x], \quad d \in \{0, 1\}$$

**Implementation:**

- In practice, $m_d(x)$ is estimated using regression methods
- Examples:
  - linear regression
  - nonparametric or machine learning models

**Connection:**

- Regression adjustment provides estimates $\hat{m}_1(x)$ and $\hat{m}_0(x)$
- These are then used in the plug-in formula for $\hat{\tau}$

**Key point:**

- Regression is not an alternative to the plug-in estimator, but a way to implement it

# Matching

**Idea:**

- Compare treated units ($D_i = 1$) with control units ($D_i = 0$) that have similar covariates $X$

**Procedure:**

- For each treated unit, find one or more control units with similar $X$
- Compare their observed outcomes

**Goal:**

- Approximate a randomized experiment by balancing covariates between treated and control groups

**Estimator:**

$$\hat{\tau}_{\text{match}} = \frac{1}{n_1} \sum_{i:\, D_i=1} \left( Y_i - Y_{j(i)} \right)$$

- $j(i)$: index of the matched control unit for treated unit $i$
- $Y_{j(i)}$: observed outcome of the matched control unit

# Matching as Conditional Comparison

**Objective:**

- Approximate the conditional expectation

$$\mathbb{E}[Y \mid D = d, X = x]$$

  by comparing units with similar covariates $X$

**Mechanism:**

- For a treated unit with covariates $X_i$, select control units with covariates $X_j$ close to $X_i$
- Use their outcomes to approximate the counterfactual outcome

**Interpretation:**

- Matching constructs local comparisons at fixed $X$
- These comparisons approximate the functions $m_1(x)$ and $m_0(x)$

**Connection:**

- Matching provides a nonparametric way to estimate the components of the plug-in estimator

# Estimating the Propensity Score

**In practice:**

- The true propensity score $e(X_i) = P(D_i = 1 \mid X_i)$ is unknown
- It must be estimated from data

**Standard approach — logistic regression:**

$$\log \frac{\hat{e}(X_i)}{1 - \hat{e}(X_i)} = \hat{\alpha} + \hat{\beta}^\top X_i$$

**Procedure:**

- Regress $D_i$ on $X_i$ using logistic regression
- Use predicted probabilities $\hat{e}(X_i)$ in place of $e(X_i)$

**Note:**

- More flexible models (random forests, neural nets) can also be used
- Misspecification of $e(X_i)$ leads to biased IPW estimates

# Inverse Probability Weighting (IPW)

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{D_i Y_i}{e(X_i)} - \frac{(1-D_i)Y_i}{1-e(X_i)}\right)$$

**Idea:**

- Give more weight to units that are underrepresented given their covariates

**Interpretation:**

- Reweight observations to create a pseudo-population
- In this pseudo-population, treatment is independent of covariates
- Thus, comparison between treated and control mimics a randomized experiment

**Note:**

- Requires overlap: $0 < e(X_i) < 1$ (otherwise weights explode)

# Why Inverse Probability Weights?

**Problem:**

- Some covariate profiles $X_i$ are more likely to receive treatment than others
- This creates imbalance between treated and control groups

**Idea:**

- Reweight observations to compensate for unequal probabilities of treatment

**Mechanism:**

- Units with low probability of their observed treatment receive higher weight
- Units with high probability receive lower weight

**Interpretation:**

- Creates a synthetic sample in which treatment assignment is independent of $X_i$

# IPW as Reweighting the Data

**Population identities:**

$$\mathbb{E}[Y(1)] = \mathbb{E}\left[\frac{D\,Y}{e(X)}\right], \qquad \mathbb{E}[Y(0)] = \mathbb{E}\left[\frac{(1-D)\,Y}{1-e(X)}\right]$$

**Implication:**

- The ATE can be written as

$$\tau = \mathbb{E}\left[\frac{D\,Y}{e(X)} - \frac{(1-D)\,Y}{1-e(X)}\right]$$

**Sample analogue:**

- Replace expectations by sample averages:

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{D_i Y_i}{e(X_i)} - \frac{(1-D_i)Y_i}{1-e(X_i)}\right)$$

**Key point:**

- The IPW estimator is the empirical version of these weighted expectations

# Augmented IPW (Doubly Robust) Estimator

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\underbrace{\hat{m}_1(X_i) - \hat{m}_0(X_i)}_{\text{Outcome Model}} + \underbrace{\frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-D_i)(Y_i - \hat{m}_0(X_i))}{1-\hat{e}(X_i)}}_{\text{Weighted Residual}}\right]$$

**Key property (double robustness):**

- Consistent if *either* $\hat{m}_d(x)$ *or* $\hat{e}(X_i)$ is correctly specified
- Does not require both models to be correct simultaneously

**Interpretation:**

- Starts from the plug-in (G-formula) estimate
- Adds IPW correction terms to remove bias from outcome model misspecification

**Why it matters:**

- In practice, neither model is perfectly specified
- AIPW provides insurance against one of the two being wrong

# From Estimation to Measurement

**So far:**

- We assumed that variables $(Y_i, D_i, X_i)$ are directly observed
- Estimation methods (regression, matching, IPW) rely on these variables

**In many applications:**

- Variables are not directly observed
- Instead, they must be extracted from raw data (e.g., text, images)

**Implication:**

- The quality of causal estimation depends on how these variables are measured

**Transition:**

- We now introduce LLMs as tools for constructing $(Y_i, D_i, X_i)$ from raw data

# Measurement with LLMs

$$\tilde{Y}_i = f_{\text{LLM}}(W_i)$$

**Notation:**

- $W_i$: raw data (e.g., text) for unit $i$
- $\tilde{Y}_i$: LLM-based measurement of the outcome
- $Y_i$: true (latent) outcome

**Key issue:**

- $\tilde{Y}_i \neq Y_i$ (measurement error)
- Errors may be systematic and depend on $W_i$ or $D_i$

**Example (congressional speeches):**

- $W_i$ = text of speech $i$
- $Y_i$ = true sentiment of the speech (latent)
- $\tilde{Y}_i$ = sentiment score assigned by the LLM
- The LLM may score speeches differently depending on party affiliation $D_i$, introducing systematic error correlated with treatment

# Latent vs Measured Variables

**Distinction:**

- $Y_i, D_i, X_i$: target variables of interest
- $\tilde{Y}_i, \tilde{D}_i, \tilde{X}_i$: variables constructed by the LLM

**In our application:**

- Only $Y_i$ is LLM-measured ($\tilde{Y}_i$); $D_i$ and $X_i$ are observed directly

**Interpretation:**

- Causal inference is defined in terms of $(Y_i, D_i, X_i)$
- In practice, we observe $(\tilde{Y}_i, \tilde{D}_i, \tilde{X}_i)$

**Implication:**

- Estimation is performed on measured variables, not on the true ones
- Discrepancies between the two induce measurement error

# Measurement Error and Bias

$$\tilde{Y}_i = Y_i + \varepsilon_i$$

**Classical case (MCAR — Missing Completely At Random):**

- $\varepsilon_i \perp (D_i, X_i)$
- Measurement error is mean-independent of treatment and covariates
- Inflates variance but does *not* bias the ATE

**LLM case (MNAR — Missing Not At Random):**

- $\varepsilon_i$ may depend on treatment $D_i$ or covariates $X_i$
- $\Rightarrow$ non-classical (systematic) measurement error
- LLMs are often MNAR because:
  - Prompt sensitivity: the LLM response may vary with the content of $W_i$, which is correlated with $D_i$ or $X_i$
  - Training bias: LLMs may have absorbed systematic biases that correlate $\varepsilon_i$ with $D_i$ or $X_i$

**Consequence:**

- $\mathbb{E}[\varepsilon_i \mid D_i] \neq 0 \;\Rightarrow$ biased estimates of treatment effects

# Implications for Estimation

- Regression estimates may be biased
- Propensity scores $e(X_i)$ may be misestimated
- Ignorability may fail after measurement

**Why:**

- Measurement error can introduce dependence between $D_i$ and potential outcomes

**Key message:**

- Measurement is part of the causal model, not a preprocessing step
- LLM outputs must be modeled, not taken as ground truth

# Preview: Correcting Measurement Error

**The problem:**

- We observe $\tilde{Y}_i$ (LLM output) instead of $Y_i$ (true outcome)
- Under MNAR, naive estimation using $\tilde{Y}_i$ yields biased causal estimates

**Correction strategy:**

- Use a small labeled dataset where both $Y_i$ and $\tilde{Y}_i$ are observed
- Use labeled data to calibrate LLM outputs
- Combine statistical correction with causal estimation
- Recover valid causal effects under assumptions

**Next step (Day 3):**

- Design-Based Supervised Learning (DSL)

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# Questions and Discussion

Thank you!

Questions?

Moses.Boudourides@northwestern.edu
Moses.Boudourides@gmail.com